github.com/benwillcutt/Markov2026

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STAT 4100
HW6

1.a) $\pi_1 = \frac{1}{2}\pi_4$, $\pi_2 = \pi_1 + \frac{1}{2}\pi_4$
$\pi_3 = \pi_2$, $\pi_4 = \pi_3$

$\pi_2 = \pi_4$, $\pi_4 = \pi_1 + \frac{1}{2}\pi_4$

$\pi_1 = \frac{1}{2}\pi_4$

so (1, 2, 2, 2), and sum is 7, so

$$\boxed{\pi = \left(\frac{1}{7}, \frac{2}{7}, \frac{2}{7}, \frac{2}{7}\right)}$$

For p(1,1): could be 1→2→3→4→1, 4 steps
could be 1→2→3→4→2→3→4→1, 7 steps
could be 1→2→3→4→2→3→4→2→3→4→1, 10 steps
13 steps, 16 steps...

n can be $4+3a+4b$ for any pos. integers a and b,
so $n = 4, 7, 8, 10, 11, 12, 13, \ldots$,

if $a=2$ and $b=0$, the $n = 4+6 = 10$ ✓
if $a=1$ and $b=1$, then $n = 4+3+4 = 11$ ✓
if $a=0$ and $b=2$, then $n = 4+8 = 12$ ✓
we can continue to add 3 to any of these to get any number ≥ 13
But there is no way to get to 9 with positive integer values (and zero) for a and b.

For p(4,4): 4→1→2→3→4, 4 steps
4→2→3→4, 3 steps
p(4,4), n can be 3, 4, 6, 7, 8, 9, 10, etc
so n can be $3a+4b$ where a and b are pos. integers and $a+b \geq 1$.

Periods:
State 1: $\gcd(4, 7, 8, \ldots) = 1$, so period = 1
State 4: $\gcd(3, 4, 6, \ldots) = 1$, so period = 1
∴ Both are aperiodic.

1.a.cont) The return time sets may be different, but their periods are the same, so this is consistent with the class property.

1.b) Going from m to 1 gives a cycle length of m
going from m to 2, then m to 1 after gives
a cycle length of $2m-2$
Continued: $3m-4$, $4m-6$, etc

so period = $\gcd(m, 2m-2, 3m-4, \text{etc})$

When m is odd, this is 1, when m is even, this is 2, so the chain is aperiodic when m is odd.

When m is odd and aperiodic, the only eigenvalue will be 1.
When m is even and the chain is periodic, so there will be 2 eigenvalues on the unit circle:
1 and -1. The -1 reflects the chains alternating behavior between 2 classes of states (b/c period = 2)

1.c) From my code, M matches Wielandt's bound of $((m-1)^2+1)$ for every m 3 to 12.
As m increases, $|R_2|$ approaches 1, starting from .7071 for m=3 to .9967 for m=12.
This means $\vec{q}_A$ takes longer to approach π as m increases.

s_i = sum of row i of A

2.a) Choose k such that r_k is the largest entry of r

$$\lambda r_k = \sum_j a_{kj} r_j \leq r_k \sum_j a_{kj} = r_k s_k$$

$$\lambda \leq s_k \leq \max_i \{s_i\}$$

So, $\lambda \leq$ the largest row sum of A ✓

Choose ℓ such that r_ℓ is the smallest entry of r

$$\lambda r_\ell = \sum_j a_{\ell j} r_j \geq r_\ell \sum_j a_{ij} = r_\ell s_\ell$$

$$\lambda \geq s_\ell \geq \min_i \{s_i\}$$

So smallest row sum of $A \leq \lambda \leq$ largest row sum of A

$$A = \begin{pmatrix} B_1 & & & \\ & B_2 & & \\ & & \ddots & \\ & & & B_8 \end{pmatrix}$$ where $B_k = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$

where a and b are the ratings of the teams results against each other.

The matrix is reducible because the blocks don't interact with each other at all.

For each block:

$$\det(B_k - \lambda I) = \begin{pmatrix} -\lambda & a \\ b & -\lambda \end{pmatrix} = \lambda^2 - ab = 0$$

$$\lambda = \pm\sqrt{ab}$$

Each block only ranks it's 2 teams, and there is no info connecting the blocks, so it can't produce a ranking of all 16 teams.

2 b) For $i \neq j$, the numerator $\sum h_{ij} + w\Pi_{ij}$ is at least $w\Pi_{ij}$ since w, Π_{ij}, and the denominator are positive

So $a_{ij} > 0$ and the diagonal entries are 0.

$(A^2)_{ij} = \sum_{k=1}^{n} a_{ik}a_{kj}$, so:

when $i=j$: $A^2_{ii} = \sum_{k \neq i} a_{ik}a_{ki} > 0$

when $i \neq j$: $A^2_{ij} = \sum a_{ik}a_{kj}$. Since $m \geq 3$, some k is not i nor is it j, so that term is a product of 2 positive numbers.

$\therefore A^2 > 0$

Since A^2 is strictly positive, there is a strictly positive eigenvector r, which is unique when normalized. Power iteration will then converge to this r from any nonnegative nonzero start.

c) Plot included at end of pdf

Prior	λ^*	$\lvert\lambda_2/\lambda^*\rvert$
Flat	.49923	.15898
Last Season	.48188	.15995
Own	.48448	.16359

For my own prior, I used the suggested one of using preseason strength estimates from 2025 results.

All of the predicted game outcomes are in my code output, but the flat prior went 5/5 while the other two priors went 4/5, incorrectly predicting VCF over OKST.

3. a) $h_n = P_n(X_T = K)$

prob p_n -> increase to $n+1$
prob q_n -> decrease to $n-1$
prob $1-p_n-q_n$ -> stay at n

$$h_n = p_n h_{n+1} + q_n h_{n-1} + (1-p_n-q_n)h_n$$

$$(p_n+q_n)h_n = p_n h_{n+1} + q_n h_{n-1}$$

so $p_n(h_{n+1}-h_n) = q_n(h_n - h_{n-1})$ ✓

Staying put at n shouldn't change the eventual hitting probability

$d_n = h_n - h_{n-1}$, so $p_n d_{n+1} = q_n d_n$ or $d_{n+1} = \frac{q_n}{p_n} d_n$

$p_0 = 1$ and $p_j = \prod_{k=1}^{j} \frac{q_k}{p_k}$

so $d_{j+1} = d_1 p_j$

$$h_n = \sum_{i=1}^{n} d_i = d_1 \sum_{j=0}^{n-1} p_i$$

$$d_1 \sum_{j=0}^{K-1} p_j = 1, \text{ so } d_1 = \frac{1}{\sum_{j=0}^{K-1} p_j}$$

$$\therefore h_n = \frac{\sum_{j=0}^{n-1} p_j}{\sum_{j=0}^{K-1} p_j}$$

For gambler's ruin, $p_0 = p$, $q_n = q$, $p_j = \left(\frac{q}{p}\right)^j$

if $p \neq q$, $h_n = \boxed{\frac{1-\left(\frac{q}{p}\right)^n}{1-\left(\frac{q}{p}\right)^K}}$ ✓ and if $p = q = 1/2$, $h_n = \boxed{\frac{n}{K}}$ ✓

3.b) $\rho_n = \beta_n(N-n)\Delta t$, $q_n = \gamma_n \Delta t$

$$\frac{q_k}{\rho_k} = \frac{\gamma k \Delta t}{\beta k(N-k)\Delta t} = \frac{\gamma}{\beta(N-k)} = \frac{N}{R_0(N-k)}$$

if $R_0 = \beta N/\gamma$

$$\boxed{P_j = \prod_{k=1}^{j} \frac{N}{R_0(N-k)}}$$

$$P_j = \left(\frac{N}{R_0}\right)^j \frac{(N-j-1)!}{(N-1)!}$$

$$h_1 = \frac{1}{\sum_{j=0}^{K-1} P_j}$$

For fixed k, $\lim_{N\to\infty} \frac{N}{R_0(N-k)} = \frac{1}{R_0}$ so $P_j = R_0^{-j}$

so $h_1 = \frac{1}{\sum_{j=0}^{K-1} R_0^{-j}}$

As $K \to \infty$:

if $R_0 < 1$, $h_1 \to 0$

if $R_0 = 1$, denominator is K, $h_1 = \frac{1}{K} \to 0$

if $R_0 > 1$, series converges to $\frac{1}{1 - 1/R_0}$

so h_1 converges to $1 - \frac{1}{R_0}$

$\therefore h_1 = \max\left\{0, 1 - \frac{1}{R_0}\right\}$

3.c) Plot included at end of pdf

d)

$$e_n = 1 + p_n e_{n+1} + q_n e_{n-1} + (1 - p_n - q_n) e_n$$

$$-q_n e_{n-1} + (p_n + q_n) e_n - p_n e_{n+1} = 1$$

From my code:

R_0	$e_1 \Delta t$
.5	1.373575
.8	1.769876
1.5	1.795853
3.0	.965486

From simulation, $R_0 = .8$ gives $e_1 \Delta t = 1.767191$, very similar to what you get when you solve the system.

$\lambda_n = \beta_n (N - n)$, $\mu_n = \gamma n$

so $p_n = \lambda_n \Delta t$, $q_n = \mu_n \Delta t$

One step equations: $\Delta t (-\mu_n e_{n-1} + (\lambda_n + \mu_n) e_n - \lambda_n e_{n+1}) = 1$

let $v_n = e_n \Delta t$, so $-\mu_n v_{n-1} + (\lambda_n + \mu_n) v_n - \lambda_n v_{n+1} = 1$

The system contains $n_0 \Delta t$, so $e_n \Delta t$ is independent of Δt.